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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations
 December 2025

AD2206-1 – NATURAL LANGUAGE PROCESSING
 (For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- What is an advantage of FastText over Word2Vec?
 - It is faster in training
 - It considers subword information, handling rare words better
 - It does not require a large dataset
 - It is a rule-based system
- Which algorithm is known for using the "Skip-gram" and "CBOW" models?
 - BERT
 - Word2Vec
 - FastText
 - GloVe
- Which of the following is NOT a neural word embedding algorithm?
 - Word2Vec
 - FastText
 - TF-IDF
 - GloVe
- A bigram model considers:
 - Only one word at a time
 - The last two words for prediction
 - The last three words for prediction
 - The entire sentence
- Which technique is used in N-gram models for predicting the next word?
 - Probabilistic estimation
 - Deep learning
 - Rule-based approach
 - Bag-of-Words model
- What is the advantage of Neural Network-based word embeddings over traditional models like BoW and TF-IDF?
 - They can capture semantic meaning and word similarity
 - They require less data
 - They do not require training
 - They are non-parametric models
- Which of the following word representation techniques considers word co-occurrence?
 - One-hot encoding
 - Bag-of-Words
 - TF-IDF
 - Word Embeddings
- What is the main limitation of the Bag-of-Words (BoW) model?
 - It does not consider the order of words
 - It requires labeled data
 - It is computationally expensive
 - It cannot handle large vocabularies
- What is the main goal of topic modeling?
 - Classify documents into predefined categories
 - Generate realistic text
 - Discover hidden themes in a collection of documents
 - Summarize long documents
- Which of the following is a common algorithm for topic modeling?
 - LDA (Latent Dirichlet Allocation)
 - TF-IDF
 - Word2Vec
 - BERT

11. What is the primary use of Graph Neural Networks (GNNs) in NLP?
 A) Capturing relationships in structured data like knowledge graphs
 B) Improving image recognition
 C) Generating synthetic speech
 D) Enhancing RNN performance
12. What is the key difference between Conditional Random Fields (CRFs) and HMMs?
 A) HMMs assume independence between output variables, while CRFs do not
 B) CRFs use generative models, while HMMs use discriminative models
 C) HMMs are deep learning models, while CRFs are statistical models
 D) CRFs cannot model sequential data
13. What is the key advantage of Transformer models over RNNs?
 A) They process sequences in parallel
 B) They use CNN layers
 C) They require less data
 D) They do not need training
14. What is the main architectural component of GPT models?
 A) CNN layers
 B) LSTM cells
 C) Transformer decoder blocks
 D) Autoencoders
15. Which task is BERT primarily designed for?
 A) Image classification
 B) Sentiment analysis
 C) Language understanding tasks
 D) Speech synthesis
16. What is the primary difference between BERT and GPT?
 A) BERT is bidirectional, while GPT is unidirectional
 B) GPT is used for classification, while BERT is used for text generation
 C) BERT requires labeled data, while GPT does not
 D) GPT uses CNNs instead of Transformers
17. Which of the following is a widely used architecture for Neural Machine Translation (NMT)?
 A) RNN
 B) Transformer
 C) CNN
 D) Naive Bayes
18. In a Question Answering (QA) system, which approach is commonly used to retrieve the answer to a question from a large text?
 A) Summarization
 B) Information retrieval
 C) Sentiment analysis
 D) Named entity recognition
19. Which of the following is a common task of Sentiment Analysis?
 A) Identifying the language of a text
 B) Detecting the positive, negative, or neutral sentiment of a text
 C) Parsing sentences for grammatical structure
 D) Identifying named entities
20. Sentiment Analysis is mainly used to determine the:
 A) Topic of a document
 B) Language of a document
 C) Emotional tone or sentiment expressed in a text
 D) Grammatical correctness of a text

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the scope and applications of computational linguistics with a suitable example.	8	L*2	1	1
	b) Apply neural network-based word embedding techniques to analyze the semantic similarity between the words "teacher" and "professor".	8	L3	2	1
2.	a) Explain collocations in NLP and discuss their importance with examples.	8	L2	1	1
	b) Explain n-gram model with an example, describing how it is used to estimate the probability of a sentence.	8	L3	2	1
3.	a) Explain the working of the TF-IDF algorithm with a mathematical formulation and an example.	8	L2	1	1

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- b) Apply n-gram models to compute the probability of the sentence "I love natural language processing" using a trigram language model. Explain the steps involved.

8 L3 2 1

Unit – II

4. a) Explain BERT's architecture, pre-training tasks and its applications in NLP tasks.

8 L2 3 1

- b) Apply Long Short-Term Memory (LSTM) networks to solve the vanishing gradient problem in NLP. Provide an example of their application.

8 L3 4 1

5. a) Compare and contrast LSTM and GRU architectures in terms of performance and efficiency.

8 L2 3 1

- b) Apply the attention mechanism in neural networks to improve machine translation. Explain its advantages over traditional models.

8 L3 4 1

6. a) Explain the architecture, working, and challenges of Recurrent Neural Networks (RNNs).

8 L2 3 1

- b) Explain the transformer networks in NLP and show how it is different from RNN – based models.

8 L3 4 1

Unit – III

7. a) Explain the approaches to Part-of-Speech (POS) tagging in NLP.

8 L2 5 1

- b) Explain the pipeline for building a question-answering system and the challenges faced in NLP applications.

8 L2 5 1

8. a) Explain extractive and abstractive text summarization techniques with examples.

8 L2 5 1

- b) Explain the architecture and workflow of a sentiment analysis system with suitable examples.

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
